NyaySetu — Project Catalogue Report

Tamper-evident evidence, summons and bail records for the criminal justice system of Maharashtra

Item	Detail
Project	NyaySetu (न्यायसेतु, "bridge of justice")
Domain	Governance and civic technology · emerging technology (blockchain, AI, GIS) · digital inclusion
State focus	Maharashtra (pilot design: Pune — Shivajinagar Police Station, Regional Forensic Science Laboratory, Sessions Court)
UN Sustainable Development Goal	SDG 16 — Peace, justice and strong institutions (targets 16.3, 16.6, 16.10)
Live system	https://nyaysetu-project.vercel.app
Public pages, no sign-in	/verify · /transparency · /handbook · /about
Smart contracts (Ethereum Sepolia, source verified)	EvidenceChain `0x383d996De8cd2a19aCC8d66CdEa2F4A3b1E7F6D4` · SummonsChain `0x18f8D239653A5542d0cBfA8d13c30E46F3c18232` · BailChain `0x90e8B17CB82848ad0fE64Eb217c495cb91E8F946`
Source code	https://github.com/SAI-RANDIVE/NyaySetu
Developed by	Sai Randive
Version	2.0 · October 2026

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1. Abstract

In a criminal case, three facts are routinely settled by somebody's word: that the evidence produced in court is the evidence that was collected, that a summons reached the person it named, and that bail conditions are being kept. Since 1 July 2024 the Bharatiya Nagarik Suraksha Sanhita (BNSS) and the Bharatiya Sakshya Adhiniyam (BSA) require audio-video recording of searches and seizures, forensic attendance at serious crime scenes, and a certificate stating the hash value of every electronic record. Police now produce far more digital evidence, and each item needs a chain of custody that a court — and the defence — can rely on.

NyaySetu computes a SHA-256 fingerprint of each piece of evidence on the officer's own device at the moment of capture, before anything is uploaded, and anchors that fingerprint to a public blockchain. Every subsequent hand-over — to the malkhana, the forensic laboratory, the prosecution and the court — re-computes the fingerprint and is refused if it differs; the refusal itself is recorded permanently. Anyone, including the accused and defence counsel, can check a file against the ledger from a public web page without an account and without trusting the police database. Summons acknowledgement and bail check-ins are recorded the same way. Personal data never goes on the ledger.

The system is deployed and working: seven role portals, a public verification service, an installable mobile web app, Marathi, Hindi and English throughout, three verified smart contracts, 183 automated tests, and measured results including 100% detection of one-byte tampering across 200 files and 100% of 258 unauthorised access attempts blocked.

2. The problem

2.1 Three questions a court should not have to take on trust

Question asked in court	How it is answered today	Consequence
Is this the evidence that was collected?	A malkhana register and case diary record that a transfer happened, not that the item which arrived is the item which left.	Chain-of-custody challenges, adjournments, acquittals on doubt; the defence cannot independently test the prosecution's copy.

Question asked in court	How it is answered today	Consequence
Did the summons reach the person named?	A process server's endorsement against a flat denial.	Repeated adjournments for service; non-appearance delays framing of charges.
Are bail conditions being kept?	Conditions written into an order nobody monitors until something has already gone wrong.	Breaches surface late; compliant accused have no way to show compliance.

2.2 Why this matters now, and why in Maharashtra

- **New statutory duties since 1 July 2024.** BNSS §105 (recording of search and seizure by audio-video means), BNSS §176(3) (forensic collection and recording at the scene for offences punishable with seven years or more), and BSA §63 (certificate for electronic records, including the hash value) turn recording and hashing into legal obligations.
- **Scale of pendency.** Maharashtra's district and taluka courts carry more than fifty lakh pending cases, the majority of them criminal (National Judicial Data Grid). Every adjournment spent disputing an exhibit or service of summons is time those cases do not get.
- **Forensic load.** Forensic laboratories report substantial backlogs, and §176(3) increases the volume of material collected at scenes. Proving that the item a laboratory examined is the item collected becomes more important, not less.

Figures change continuously; current values should be read from NJDG (njdg.ecourts.gov.in) and the Directorate of Forensic Science Laboratories, Maharashtra.

2.3 A recent case: the swapped blood sample (Pune, May 2024)

In the early hours of 19 May 2024 a car driven by a minor killed two young people in Kalyani Nagar, Pune. A blood sample was drawn at Sassoon General Hospital to test for alcohol. According to the prosecution, the sample was discarded and replaced with another person's blood — DNA later matched it to the driver's mother — before it reached the laboratory. The substitution came to light only because police, acting on information about possible tampering, had a second sample drawn at the District Hospital, Aundh, and sent both for DNA analysis. Two doctors and a hospital staff member were arrested, nine people were held by August 2024, and the matter is before the court. (Sources: The Tribune, Scroll, The Week/PTI; listed in §20.)

The lesson is structural. Nothing tied the vial to the person, its seal and the minute it was drawn, and nobody downstream had anything to check it against. Detection depended on a tip-off and a second sample — luck, not system.

With NyaySetu. The officer records a short collection clip and binds the serial of the vial's tamper-evident seal to the exhibit; both are anchored within minutes. A substitute vial must carry a different seal. When the laboratory types the serial it sees, the mismatch refuses the intake, the wrong reading is itself anchored, and the prosecutor and the court are alerted — the same day, without DNA. The interactive replay is at https://nyaysetu-project.vercel.app/case-study.

What NyaySetu would not have prevented: if the people drawing the sample put the wrong person's blood into the original sealed vial in front of the camera, no ledger could know. The clip makes that far harder to do unseen.

2.4 Who is affected

Stakeholder	Pain today	What changes with NyaySetu
Investigating officer, malkhana in-charge	Paper registers; §63 certificates typed by hand; video handled on personal phones	Capture on a phone with an offline queue, digital malkhana, certificate generated from the record
Forensic laboratory	No proof that what arrived is what was sent	Re-hash on arrival; a mismatch is refused, anchored and alerted
Public prosecutor	Discovers custody gaps late	Readiness view per case; alerts for every refused hand-over
Judge and court registry	Disputes about exhibits and service; unmonitored bail	One verifiable record; summons status from the ledger; ranked bail compliance with a map
Defence counsel, legal aid	Must trust the prosecution's copy and log	Independent verification against a public ledger
Accused, witnesses (often first-generation litigants)	Obligations in legal English; no way to show compliance	Plain-language explanations in Marathi, Hindi or English, read aloud; their own check-in record
The public	Little visibility into how evidence is handled	A transparency page with live counts and every ledger transaction

3. Objectives

1. Make the integrity of each item of digital evidence **verifiable by anyone**, from capture to courtroom, without trusting any single party's database.
2. Record **every custody hand-over with a re-check**, and make a failed check impossible to hide or quietly correct.
3. Generate the **BSA §63 certificate** and custody documents from the record itself, each with a QR code back to public verification.
4. Record **summons acknowledgement** as a timestamped, authenticated event.
5. Monitor **bail conditions** as code, with geo-fenced check-ins whose verdict is computed by the contract and reproducible by anyone.
6. Keep **personal data off the public ledger** entirely.
7. Serve people who are not lawyers: **Marathi, Hindi and English**, read-aloud, large text, low-end Android, intermittent connectivity.
8. Be **deployable now** at negligible cost, with a clear path to state scale.

4. What exists today, and the gap

Existing system	What it does well	What it does not do	How NyaySetu relates
e-Sakshya (Ministry of Home Affairs / NIC)	Records §105 search-and-seizure media and links it to the FIR	Verification relies on the government's own copy; no public, independent check; does not follow the item through laboratory, malkhana and court; no summons or bail	**Complements it.** NyaySetu can anchor the fingerprint of every e-Sakshya recording and verify it at each later hand-over.
CCTNS / ICJS	Police case records and data exchange across police, courts, prisons, forensics and prosecution	No tamper-evident fingerprint per exhibit that the defence can check	NyaySetu adds an integrity layer; the exchange with ICJS is designed (`docs/INTEGRATION.md`) but not yet connected.
e-Courts / NJDG	Case status, cause lists, pendency statistics	No evidence-integrity or bail-compliance layer	Summons and bail status from NyaySetu can feed case status.
Department of Justice "Nyaya Setu" chatbot	General legal information on WhatsApp	Not case-specific; no evidence, summons or bail record	Different purpose; NyaySetu is a record system, not an information service.
Commercial digital evidence management systems	Storage, access control, audit logs	Audit log is held by the same party whose conduct is in question	NyaySetu's proof is on a ledger no party in the case controls.

The gap: no existing Indian system lets a party who does not trust the prosecution verify, independently and without permission, that an exhibit is unchanged since capture — and none ties evidence custody, service of summons and bail compliance into one verifiable record.

5. The solution

5.1 The one idea

A SHA-256 fingerprint is a 64-character value computed from a file. Change one bit of the file and the fingerprint changes completely; it cannot be reversed to recover the file. NyaySetu computes the fingerprint **on the device that captured the evidence, before upload**, writes it to a public blockchain, and compares against it at every later step. Anyone holding the file can recompute the fingerprint and compare it with the ledger. They need to trust mathematics and a public ledger — not the police, the laboratory or the court's database.

5.2 Five steps, each one checkable

1. **Capture** — photo, video or file on the officer's phone or browser; the fingerprint, time, GPS fix and accuracy are recorded before upload. Without signal, the capture waits safely in an offline queue and syncs later without duplicates.
2. **Anchor** — the fingerprint of a capture manifest is written to the EvidenceChain contract. The record shows honestly where it has got to: saved locally, submitted, confirmed on chain, review required, or integrity mismatch.
3. **Hand over** — malkhana, laboratory, prosecution and court each re-hash what they received before accepting it. A mismatch is anchored on chain, the transfer is refused, and an alert is raised to the prosecution and the court.
4. **Verify** — anyone can open the item's public Integrity Passport (by reference or QR code), or simply drop the file on the verification page, and see whether it matches the ledger.
5. **Produce in court** — the BSA §63 certificate, seizure memo and custody record are generated from the record, each carrying the SHA-256 digest and a QR code to public verification.

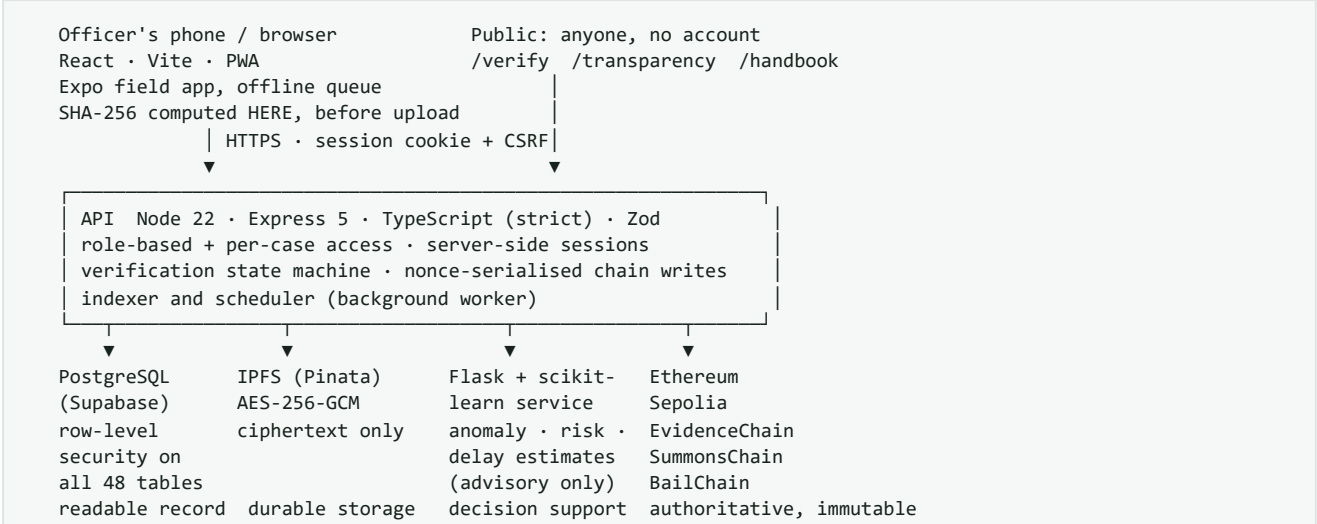
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6. What is new

1. **Physical exhibits are covered, not only files.** The serial of a tamper-evident seal is bound to the exhibit and checked at every hand-over, so a substituted sample is refused at the first one.
2. **Fingerprint at the point of capture, on the officer's device.** The proof starts before the file reaches any server, so no server — including NyaySetu's own — can substitute a file unnoticed.
3. **Verification without permission.** A public page, with no account, lets the defence, the court or the accused check a file against the ledger. A file can be checked even without its reference: the system finds the matching record from the fingerprint alone.
4. **A failed check is permanent.** Integrity mismatches are written to the ledger, not to an editable log, and the hand-over is refused.
5. **Evidence, summons and bail in one verifiable record**, with three purpose-built contracts.
6. **Bail compliance computed on chain.** Geo-fence arithmetic runs inside the contract using integer micro-degrees, so the verdict on each check-in is reproducible by anyone and cannot be altered by the operator.
7. **Paper that verifies itself.** Every certificate and custody record carries a QR code that opens the public verification page for that item.
8. **Designed for the accused, not only for officials.** Obligations explained in Marathi, Hindi and English, read aloud by the phone, with a way to give reasons for a missed check-in.
9. **Radical, privacy-preserving transparency.** A public page shows live counts and every recent ledger transaction, with no personal data, and a "tamper lab" that lets anyone see for themselves what a one-bit change does.
10. **Honesty built into the interface.** The system states what it does not prove (a matching fingerprint shows a file is unchanged, not that its content is true), labels AI output as decision support, and labels simulated components as simulated.

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7. Architecture



Principle: the ledger holds only what must be impossible to revise — a fingerprint, an acknowledgement, a condition, a verdict. Everything readable lives in PostgreSQL. An indexer reconciles the two; where they disagree, the ledger is right and the disagreement is shown.

Layer	Technology	Why
Smart contracts	Solidity 0.8.24, OpenZeppelin 5 (AccessControl, ReentrancyGuard), Hardhat	Role-gated writes; built-in overflow checks; 53 tests on a real EVM
API	Node 22, Express 5, TypeScript strict, Zod validation, Ethers v6	Every request body parsed, not cast; one module owns every chain call
Database	PostgreSQL (Supabase), 13 migrations, 48 tables, row-level security on every table, audit triggers	Least privilege enforced by the database as well as the API
Storage	IPFS via Pinata; AES-256-GCM before upload	The store never sees plaintext and is never trusted for integrity
Web	React 19, Vite, Tailwind, WebCrypto, installable PWA, Leaflet + OpenStreetMap	Fingerprint in the browser; works on low-end Android; no map API key
Mobile	React Native (Expo), expo-crypto, offline queue with idempotency keys	Field capture without signal, never duplicated
Intelligence	Python, Flask, scikit-learn (Isolation Forest, Random Forest, Gradient Boosting)	Transparent, explainable models, labelled as advisory
Hosting	Vercel (web, API and AI as services; background worker as a daemon), Supabase, Pinata	Deployed today; all components replaceable by State data-centre equivalents

8. Modules and features

8.1 Evidence (SaakshyaSetu · EvidenceChain)

- Case registration (with or without an FIR); evidence capture with GPS accuracy, device and time; resumable chunked upload for large video.
- Seizure memos, digital malkhana register with movements, laboratory intake with seal condition, forensic reports with versioned corrections.
- Custody transfer with mandatory re-hash; mismatch refused and anchored.
- Verification states shown on every item, with state history.
- Integrity alerts to prosecution and court; two-person approval for sensitive actions.

8.1a Physical exhibits — seal binding

A fingerprint proves a file is unchanged; a blood sample, a weapon or a packet of narcotics is not a file. What travels with a physical exhibit is a tamper-evident seal with a printed serial.

- The seizing officer binds the serial to the exhibit once, before the first hand-over. The serial stays off chain; a digest salted with the internal exhibit identifier is anchored.
- Every receiving party (laboratory intake today; custody, malkhana and court production use the same check) types the serial it actually sees. Serials are compared ignoring case, spaces and dashes.
- A different serial refuses the intake, anchors the wrong reading as its own record, and raises a serious `SEAL_MISMATCH` alert to the prosecution and the court. A seal cannot be re-bound quietly.
- Implemented in migration `014_exhibit_seals.sql`, `seal.service.ts`, and tested in `backend/test/f21-exhibit-seals.test.ts`.

8.2 Summons (SammansSetu · SummonsChain)

- Issue a summons with a 72-hour acknowledgement window the contract enforces.
- Acknowledgement by one-time code with time, coordinates and a device fingerprint, recorded on chain; service failure is detected automatically when the window lapses.

8.3 Bail (JaminSetu · BailChain)

- Conditions encoded on chain; geo-fence and check-in interval configured by the judge.
- Check-ins verified by one-time code; the contract decides inside or outside the fence; missed check-ins flagged by the scheduler.
- Ranked compliance board for the judge with reasons, the accused's explanations, and a map of the geo-fence and every check-in.

8.4 Public services (no account)

- **Integrity Passport** per item: digests, transaction, custody stages; PDF and QR code. Never names, places or case references.
- **Verify by file**: drop a file without knowing its reference; the system finds its record from the fingerprint and confirms against the chain.
- **Transparency page**: live counts, contract and transaction links, measured cost per operation, and the tamper lab.

- **Handbook**: the whole system explained for every role, publicly.

8.5 Documents

- BSA §63 certificate (Part A and Part B) generated from the record, with the anchored SHA-256 digest and a QR code; editable template managed by the court registry with a fixed disclaimer.
- Seizure memo, custody record, case final report and Integrity Passport PDFs. Samples: `docs/samples/`.

8.6 Governance and administration

- Accounts only by invitation from the court registry; no public sign-up.
- Password plus emailed one-time code; sessions held server-side and revocable instantly.
- Per-case assignment; access requests from defence approved by the court; complete audit trail of actions and refusals; chain health and keeper balance for the registry.

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9. End-to-end workflow

#	Actor	Action	What is recorded on chain
1	Police	Registers a case; captures a photo at the scene (offline if needed)	Salted case key; fingerprint of the capture manifest
2	Police	Records the seizure memo and malkhana entry	Fingerprints of memo and movement
3	Laboratory	Receives the item, re-hashes, records seal condition	Custody transfer with confirmed digest
3a	Laboratory	Receives a **different** file	Integrity mismatch, permanently; transfer refused; alert raised
4	Laboratory	Files the forensic report; a later correction keeps both versions	Report fingerprints
5	Prosecution, court	Accept custody with re-check	Custody transfers
6	Defence	Requests access; on approval verifies files independently	Integrity check result
7	Court	Issues summons	Summons with 72-hour window
8	Accused	Acknowledges with a one-time code	Acknowledgement with time, coordinates, device
9	Court	Grants bail with conditions and a geo-fence	Conditions, fence, interval
10	Accused	Checks in; one inside the fence, one outside	Each check-in and the contract's verdict
11	Court registry	Generates the §63 certificate and final report	— (documents carry the anchored digests and QR codes)

#	Actor	Action	What is recorded on chain
12	Anyone	Scans the QR, or drops the file on /verify	— (reads the chain)

The complete sequence is automated end to end over HTTP by `npm run flow:check` (and `npm run demo:data`, which leaves the record in place).

10. Alignment with the new criminal laws

Provision	Requirement	How NyaySetu supports it
BNSS §105	Search and seizure recorded through audio-video electronic means, preferably a mobile phone	Capture on the phone with fingerprint, time and location before upload; offline queue for areas without signal
BNSS §176(3)	Forensic expert to visit the scene of offences punishable with 7+ years, collect evidence and record the process	Laboratory intake with re-hash and seal condition; custody refused on mismatch
BSA §63	Certificate accompanying electronic records, including hash value(s)	Certificate generated from the record with the SHA-256 digest anchored on chain, QR-verifiable
Supreme Court, 22 May 2026	*Pune Bar Association v. Union of India*, 2026 SCC OnLine SC 1297: §63(4) upheld; a hash value is "synonymous with an electronic fingerprint"	The fingerprint is computed at the scene and anyone can check it, free
BNSS §§ 63–71 (summons)	Service of summons, including electronic service	Authenticated, timestamped acknowledgement recorded on chain
BNSS bail provisions	Conditions imposed by the court	Conditions as code; compliance computed and ranked for the court

What is not claimed. NyaySetu does not decide admissibility. The certificate and forms are templates filled from the record, to be checked by a legal practitioner. The ledger is supporting proof of integrity, not a substitute for the certificate or for the court's assessment.

11. Security and privacy

11.1 What goes on the public ledger, and what never does

On chain	Off chain
A salted case key (neither FIR nor case reference recoverable)	Evidence files, encrypted with AES-256-GCM before storage
Fingerprint of each item's capture manifest	Names, FIR details, addresses, witnesses

On chain	Off chain
Custody transfers and integrity checks, including failures	Exact evidence coordinates (only in the manifest, readable by assigned roles)
Fingerprints of seizure memos, malkhana movements, reports	Report text, court orders, comments and notes
Summons issue and acknowledgement (with acknowledgement coordinates)	Accounts, sessions, access rules
Bail conditions and check-ins with their coordinates (needed for the contract to compute the fence)	

Aadhaar numbers are never stored; they are converted to keyed HMAC tokens, and the table holding tokens is unreachable through the API.

11.2 Controls

- Server-side sessions (revocable at once), CSRF protection, strict per-route rate limits, password plus emailed one-time code.
- Role-based access and per-case assignment enforced by the API **and** by row-level security in the database; an authorisation test matrix covers every route.
- Every chain write serialised through a nonce ledger; every action and every refusal in an audit log.
- Public endpoints return only what the privacy rule allows; a test asserts the transparency figures contain no case reference, identifier, digest or name.
- Security review documented in `docs/SECURITY.md`.

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12. Artificial intelligence, used carefully

Model	Purpose	Safeguard
Evidence anomaly screening (Isolation Forest + explainable rules)	Flags implausible captures: GPS accuracy, clock skew, mocked location, unusual sizes	Rules always explain the flag; the model only adds outliers for human review
Bail risk banding (Random Forest)	Supports the court's monitoring intensity	Labelled "decision support, never a finding"; never decides
Case delay estimate (Gradient Boosting)	Indicates likely delay with an interval	Trained on declared distributions, documented in full
Statute reference	All 358 sections of the Bharatiya Nyaya Sanhita, 2023	Severity treated as a lower bound where the source omits sub-sections

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13. Inclusion and accessibility

- **Languages:** the interface in English, हिंदी and मराठी; plain-language explanations for the accused.

- **Read-aloud:** explanations spoken by the phone's own speech engine (Marathi → Hindi → Indian English fallback); nothing leaves the device.
- **Low-end devices:** installable web app; main bundle about 60 KB compressed; maps and animation loaded only when needed.
- **Connectivity:** offline capture queue that never duplicates.
- **Accessibility:** large-text mode, light and dark themes, keyboard navigation, skip links; every page tested with axe-core for WCAG violations.
- **Free to the user:** no app store, no account for verification, no paid map or speech API.

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14. Quality: how it is verified

14.1 Automated tests

Suite	Count	What it covers
Smart contracts (Hardhat)	53	Roles, custody stages, mismatch anchoring, summons window, geo-fence arithmetic, breaches
API (node:test, over real HTTP)	121	Every feature, an authorisation matrix over the API's routes, rate limits, PDFs, public privacy rules
Mobile	9	Offline queue: ordering, idempotency, retry
UI smoke and accessibility (Puppeteer + axe-core)	44 pages	Every page of every role, desktop and mobile, no console errors, no WCAG violations

14.2 Measured results

Metric	Result
Tampered files detected (one byte changed, 200 files)	**100%**
Genuine files confirmed (no false alarms)	**100%**
Unauthorised read attempts blocked (258 attempts, six roles)	**100%**
Items lost with 35% of chain writes failing	**0** (all confirmed after one retry pass)
Public check latency, in process (p50 / p95)	3.3 ms / 4.9 ms

Method and limits: `docs/metrics/METRICS.md`.

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15. Deployment

Live today at https://nyaysetu-project.vercel.app, with the health of every subsystem published at `/api/health` and summarised in the header of every public page.

Component	Where it runs now	State deployment equivalent
Web app	Vercel (static, global CDN)	State Data Centre / MeghRaj web tier
API	Vercel serverless functions (Node 22)	Containers on State cloud
Background worker (indexer, scheduler)	Vercel daemon (container)	Same container image
AI service	Vercel Python service	Same container image
Database	Supabase PostgreSQL (Singapore region)	PostgreSQL in an Indian data centre
File storage	IPFS via Pinata (encrypted)	Private IPFS cluster or object storage in India
Ledger	Ethereum Sepolia (public testnet)	Layer-2 network, or a permissioned EVM ledger (e.g. Hyperledger Besu) run by State and judiciary nodes
Email	Brevo SMTP	NIC email relay

Everything is configured by environment variables (`.env.hosting.example`), and the same code runs on a laptop with no accounts at all (`npm run local`: a local chain, in-memory database and the web app). Step-by-step hosting guides are in `docs/hosting.md` and `docs/production.md`.

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16. Feasibility

16.1 Technical

Working end to end today, on public infrastructure, with the measured results above. Every piece is open-source and replaceable. The contracts are standard Solidity and run unchanged on any EVM network.

16.2 Cost of the ledger — measured, not estimated

Gas used per operation, measured with `npm --prefix contracts run gas`:

Operation	Gas used
Register an item of evidence	3,60,093
Custody transfer with re-check	1,56,407
Issue a summons	2,40,107
Acknowledge a summons (with location and device)	1,31,376
Set bail conditions	2,76,884
Configure bail monitoring (geo-fence)	1,03,946
Bail check-in (fence computed on chain)	2,35,892

One item followed from scene to court (registration plus three custody transfers) uses about 8.3 lakh gas. The cost is independent of file size, since only a 32-byte fingerprint is written.

- **Public testnet (prototype):** free.
- **Ethereum mainnet:** at 10 gwei, about 0.008 ETH per item from scene to court — too expensive at state scale, and not proposed.
- **Layer-2 rollup:** the same contracts at a small fraction of mainnet cost, with security inherited from Ethereum.
- **Permissioned EVM ledger (recommended for the State):** no gas fees; the cost is that of running validator nodes, which can be held by independent institutions (police, judiciary, forensic directorate, a legal-services authority) so that no single party controls the record.

16.3 Operational

- Officers use the phone they already carry; capture takes the same steps as a photo, plus an automatic fingerprint.
- No change to who holds the evidence; NyaySetu adds a check at each existing hand-over.
- The court registry controls accounts; there is no public sign-up to police.
- A role-by-role handbook is public, and every screen explains its state in plain language.

16.4 Legal

Aligned with BNSS §105, §176(3) and BSA §63 as above. The ledger stores no personal data, which keeps it compatible with the Digital Personal Data Protection Act, 2023: personal data stays in an access-controlled database that can be corrected or erased, while the ledger holds only fingerprints and keys that do not identify anyone.

16.5 Scalability

- The API is stateless and scales horizontally; the database is standard PostgreSQL with indexed access paths.
- Chain writes are queued and serialised per signer; throughput can be raised by adding signers, by batching many fingerprints into one Merkle root per block interval, or by moving to a layer-2 or permissioned network.
- At Maharashtra's scale (lakhs of items per year) the database load is modest; the ledger is the constraint, which is why §16.2 recommends a permissioned EVM network for state rollout.

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17. Impact

SDG 16 target	Contribution
16.3 Equal access to justice	The defence and the accused can verify evidence and see their obligations in their own language
16.6 Effective, accountable, transparent institutions	Every hand-over is checked; failures cannot be hidden; public transparency page

SDG 16 target	Contribution
16.10 Public access to information	Verification and aggregate figures available to anyone, with no personal data

Indicators to measure in a pilot: share of exhibits with a complete, verified chain of custody; number of custody disputes raised in court; adjournments attributed to service of summons; time from seizure to laboratory intake; share of missed bail check-ins detected within one day; time to produce a §63 certificate.

18. Roadmap and pilot plan

Phase	Scope	Duration
1. Pilot	One police station (Shivajinagar, Pune), the regional forensic laboratory, one sessions court, one legal-aid clinic	3 months
2. District	All police stations of one district; malkhana and laboratory workflows	6 months
3. State	Integration with CCTNS / ICJS and e-Courts; permissioned ledger with independent validator nodes	12 months

Next technical steps: device-signed mobile captures; anchoring of e-Sakshya recordings; licensed Aadhaar authentication in place of the sandbox provider; Bhashini integration for voice in more Indian languages; Merkle batching of anchors; review of Hindi and Marathi text by native speakers.

19. Limitations

- A matching fingerprint proves a file is unchanged since it was anchored. It does not prove that what the file shows is true, or how it was collected.
- If a device is compromised before the fingerprint is computed, the proof starts from the compromised file. GPS accuracy, device time and clock-skew checks flag anomalies; the court still weighs the evidence.
- GPS can be wrong or spoofed; accuracy is recorded with every fix, and mocked locations are flagged.
- AI output is decision support and never decides anything.
- Citizen one-time codes use a sandbox provider; real Aadhaar authentication requires a licensed authentication user agency.
- Bail check-in coordinates are on chain because the contract computes the fence; moving that computation to a zero-knowledge proof is future work.
- The prototype ledger is a public testnet; production needs the network in §16.2.

20. References

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8. NIST FIPS 180-4, Secure Hash Standard (SHA-256).
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